# Monish Suresh

M.Sc. Student

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## PROFILE

Information Engineering and Computer Science student with experience across embedded firmware development (STM32, ESP32, Arduino) and PCB production (pick-and-place machine, KiCad, soldering).

## Portfolio Link

https://monishsuresh.github.io/

## WORK EXPERIENCE

### Student Worker, RF-Frontend

*November 2022 – present | Kamp-Lintfort, Germany*

- Contributed to STM32 firmware development by expanding Modbus command features, adding firmware update options, and fixing bugs
- Implemented simulations calculating s-parameters and farfield of cylinder dipole antenna using Python and MEEP (an electromagnetic simulation package), comparison of the results with CST studio simulations for validation
- Contributed to the frontend development of an inventory management system using the Angular framework
- Expanded a C#/.NET voltage probe control application to support a new voltage probe variant and fixing existing bugs
- Worked on PCB production performing tasks such as operating a pick-and-place machine, solder paste application, and microscope-based inspection of PCBs
- Performed manual soldering of SMD components, including 0201 packages, and connectors on PCBs
- Performed testing and tuning of produced PCBs to ensure quality
- Participated in weekly team meetings and provided updates on project progress

### Intern, FabLab Kamp-Lintfort

*May 2022 – October 2022 | Kamp-Lintfort, Germany*

- Developed firmware for an ESP32-based IoT system to monitor and control a hydroponic farm, integrating EC, pH, temperature, humidity, and light sensors alongside an RTC module
- Implemented closed-loop control logic on the ESP32, driving pumps and valves via MOSFET switching based on live EC, pH, and timing conditions to automate the water-quality check cycle
- Built and iterated a ThingsBoard dashboard connected to the ESP32, streaming sensor data in time-series and adding interactive controls to trigger pump/valve actions and LED toggling
- Designed a hydroponic desktop farming system in Fusion360, including tank, sensor clamps, and mounting hardware, and produced physical parts via 3D printing
- Designed and soldered custom perfboard and PCB circuits (KiCad)

## EDUCATION

### Hochschule Rhein Waal

- M.Sc. in Information Engineering and Computer Science, October 2025 – present
- B.Sc. in Communication and Information Engineering, October 2019 – August 2025

## SKILLS

Programming Languages: Python, C, C#

Embedded/Firmware: STM32, FreeRTOS, ESP32, Arduino

Tools & Platforms: Git, STM32Cube, MATLAB, Simulink, KiCad, LTspice

Languages: English (fluent), German (B2.1), Tamil, Hindi